

# Danielle A. Ripsman

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 <https://danirip.github.io>

 Google Scholar

 [github/DaniRip](https://github.com/DaniRip)

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**Research Summary:** Large-scale and robust optimization for healthcare decisions; learning optimization models and algorithms from data using mixed-integer, Pareto-robust and inverse optimization methods; applications in radiation therapy planning, home care scheduling and the adoption of AI tools in patient-facing care.

## ■ ACADEMIC APPOINTMENTS

|                                                   |                                                                                                                                                                                                                                                                                                                                 |
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| <b>Johns Hopkins University</b><br>2025 – Present | <b>Malone Postdoctoral Fellow, Malone Center for Engineering in Healthcare</b><br>Whiting School of Engineering, Baltimore, MD, USA<br><b>Additional Affiliations:</b> Center for Systems Science and Engineering (CSSE), Hopkins Business of Health Initiative (HBHI)<br><b>Mentors:</b> Dr. Kimia Ghobadi and Dr. Todd McNutt |
| <b>University of Toronto</b><br>2024 – 2025       | <b>Postdoctoral Fellow in Operations Management &amp; Statistics</b><br>Rotman School of Management, Toronto, ON, Canada<br><b>Additional Affiliations:</b> Schulich School of Business, York University; Department of Management, University of Toronto Scarborough<br><b>Mentors:</b> Dr. Andre Cire and Dr. Adam Diamant    |

## ■ EDUCATION

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|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>University of Waterloo</b><br>2018 – 2024 | <b>Ph.D. in Management Sciences</b><br>Management Science & Engineering Department, Waterloo, ON, Canada<br><b>Advisor:</b> Dr. Houra Mahmoudzadeh<br><b>Thesis:</b> Large-Scale Optimization Approaches for Radiation Therapy                                          |
| <b>University of Waterloo</b><br>2016 – 2018 | <b>M.A.Sc. in Management Sciences</b><br>Management Sciences Department, Waterloo, ON, Canada<br><b>Advisor:</b> Dr. Houra Mahmoudzadeh<br><b>Thesis:</b> Robust Direct Aperture Optimization Methods for Cardiac Sparing in Left-Sided Breast Cancer Radiation Therapy |
| <b>University of Toronto</b><br>2010 – 2016  | <b>B.A.Sc. in Industrial Engineering</b><br>Mechanical & Industrial Engineering Department, Toronto, ON, Canada<br>Graduated with Honours and an Engineering Business Minor                                                                                             |

## ■ PUBLICATIONS

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| <b>Refereed Journal Articles</b> | <ol style="list-style-type: none"><li>1. <b>D. A. Ripsman</b>, S. H. Hristov, G. Gola, W. Kwong, M. Singh, E. Osei, J. Darko and H. Mahmoudzadeh. "A comparison of voxel sampling approaches for intensity-modulated radiation therapy." <i>Medical Physics</i>, 52(12):e70179, 2025. [Link]</li><li>2. <b>D. A. Ripsman</b>, F. Rahimi, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Light Pareto robust optimization for IMRT treatment planning." <i>Medical Physics</i>, 50(5):2637–2648, 2023. [Link]</li><li>3. <b>D. A. Ripsman</b>, T. G. Purdie, T. C. Y. Chan and H. Mahmoudzadeh. "Robust direct aperture optimization for radiation therapy treatment planning." <i>INFORMS Journal on Computing</i>, 34(4):2017–2038, 2022. [Link]</li><li>4. <b>D. A. Ripsman</b>, D. M. Aleman and K. Ghobadi. "Interactive visual guidance for automated stereotactic radiosurgery treatment planning." <i>Expert Systems with Applications</i>, 42(21):8337–8348, 2015. [Link]</li></ol> |
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**Under Review** 1. F. Rahimi, **D. A. Ripsman**, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Superior light Pareto robust optimization with application to radiation therapy." *Manufacturing & Service Operations Management*. **Resubmitted following a major revision; under review.**

**Working Papers** 1. **D. A. Ripsman\***, R. Chiplunkar\* and K. Ghobadi. "Early reach and workload associations of an ambient AI scribe in community primary care." Manuscript in preparation. (\*Co-first authors.)

2. F. Parker, **D. A. Ripsman**, E. Springer and K. Ghobadi. "Pivoting under partial observability: Exact search, learned policies, and degeneracy in simplex pivot selection." Manuscript in preparation.

3. **D. A. Ripsman**, A. Cire and A. Diamant. "A decision diagram approach to multi-period home care scheduling under uncertainty." Manuscript in preparation.

4. **D. A. Ripsman**, S. A. Alev and H. Mahmoudzadeh. "A geometric approach to beam angle selection in radiation therapy treatment planning." Manuscript in preparation.

5. **D. A. Ripsman** and H. Mahmoudzadeh. "Closed-form constraint generation for classes of robust polyhedral uncertainty." Manuscript in preparation.

## ■ PRESENTATIONS

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**Conference Presentations** 1. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload: A multi-site DiD cohort study", INFORMS Healthcare, Raleigh, NC, July 2026.

2. **D. A. Ripsman** and K. Ghobadi "Generative Inverse Optimization for Healthcare Decision Making", INFORMS Healthcare, Raleigh, NC, July 2026.

3. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload", CORS Annual Meeting, Kingston, ON, June 2026.

4. **D. A. Ripsman**, A. Cire and A. Diamant. "Multi-Period Home Care Scheduling Under Uncertainty", CORS Annual Meeting, Edmonton, AB, June 2025.

5. **D. A. Ripsman**, "A Geometrical Approach to Beam Angle Optimization", ICS Conference, Toronto, ON, March 2025.

6. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi & H. Mahmoudzadeh, "Light Pareto Robust Optimization for IMRT Treatment Planning", CORS Annual Meeting, London, ON, June 2024.

7. **D. A. Ripsman**, "Integrating Uncertainty Mitigation into Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Phoenix, AZ, October 2023.

8. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Healthcare, Toronto, ON, July 2023.

9. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", CORS/Optimization Days, HEC Montréal, QC, May 2023.

10. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Indianapolis, IN, October 2022.

11. **D. A. Ripsman**, H. Mahmoudzadeh, "Structural Solution Approaches to Robust Optimization with Polyhedral Uncertainty", CORS/INFORMS International, Vancouver, BC, June 2022.

12. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "Closed-form Constraint Generation for Classes of Robust Polyhedral Uncertainty", INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.

13. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
  14. **D. A. Ripsman**, H. Mahmoudzadeh, "Closed-form constraint generation for robust radiation therapy treatment planning", CORS Annual Meeting, (Virtual), May 2021.
  15. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning", INFORMS Healthcare, Cambridge, MA, July 2019.
  16. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning", Canadian Healthcare Optimization Workshop, Saskatoon, SK, May 2019.
  17. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning", CORS Annual Conference, Halifax, NS, June 2018.
  18. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Mixed Integer Optimization for Radiation Therapy Treatment Planning with Delivery Constraints", Optimization Days, Montreal, QC, May 2017.
  19. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. "Interactive visual decision support for radiation therapy treatment planning", Industrial & Systems Engineering Research Conference (ISERC), Montreal, QC, June 2014.
  20. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. "Interactive visual decision support for radiation therapy treatment planning", Undergraduate Engineering Research Day (UnERD), University of Toronto, ON, August 2013.
- Second place in the Industrial Engineering Oral Presentations category.**

- Poster Presentations**
1. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. "The Impact of AI Scribe on Primary Care Workload", Department of Medicine and Whiting School of Engineering Research Retreat, Baltimore, USA, 2026.
  2. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. "The Impact of AI Scribe on Primary Care Workload", Johns Hopkins Research Symposium on Engineering in Healthcare, Baltimore, USA, 2025.
  3. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. "Graphical Interface for Interactive Parameter Selection in Radiation Therapy Treatment Planning", MIE Research Symposium, University of Toronto, Toronto, Canada, June 2013.

## ■ AWARDS & FELLOWSHIPS

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### Fellowships & Scholarships

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|-----------------------|---------------------------------------------------------------------------------------------------------|
| <i>2025 – Present</i> | ■ Malone Postdoctoral Fellowship, Malone Center for Engineering in Healthcare, Johns Hopkins University |
| <i>2019 – 2022</i>    | ■ NSERC Canada Graduate Scholarship–Doctoral (CGS-D) (\$105,000; \$35,000 for 3 years)                  |
| <i>2018 – 2022</i>    | ■ Engineering Excellence PhD Fellowship (\$120,000; \$30,000 for 4 years)                               |
| <i>2018</i>           | ■ Provost's Doctoral Entrance Award for Women (\$5,000)                                                 |
| <i>2017</i>           | ■ University of Waterloo Graduate Scholarship (\$750)                                                   |
| <i>2013</i>           | ■ NSERC Undergraduate Student Research Award (\$4,500)                                                  |
| <i>2010 – 2011</i>    | ■ John Hirschorn Memorial Scholarship (\$20,000; \$10,000 for 2 years)                                  |
| <i>2010</i>           | ■ University of Toronto Scholars Scholarship (\$5,000)                                                  |

### Teaching Awards

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| <i>2020, 2022</i> | ■ Sandford Fleming Foundation Teaching Assistant Award, for the R laboratory of the introductory statistics course, University of Waterloo |
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## Competitions & Honours

- 2024 ■ **Finalist** (Health Care Operations Research SIG and Open categories), CORS Student Paper Competition, London, ON — “Superior and light Pareto robust optimization”
- 2019 ■ **Winner**, CORS Health Care Operational Research SIG Student Presentation Competition, Saskatoon, SK
- 2018 ■ **Finalist**, Three Minute Thesis (3MT), University of Waterloo — first place in the departmental heat
- 2014 ■ Third place, Simulation Competition, IIE National Student Conference, Toronto, ON
- 2013 ■ Second place, Industrial Engineering Oral Presentations, Undergraduate Engineering Research Day, University of Toronto, ON
- 2013 ■ Third place, Operations Research Case Competition, Industrial Engineering Competitions Day, University of Toronto, ON
- 2012 ■ First place, Student Design Competition, American Society of Mechanical Engineers, University of Toronto, ON

## ■ GRANTS & FUNDING

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**NIH R01 (NHLBI) SCH: Intelligent Decision-Support System  
for ICU Respiratory Management**

2026 – 2030

*R01HL192731, NSF–NIH Smart and Connected Health program. PIs: Dr. James Fackler, Dr. Kimia Ghobadi and Dr. Chien-Ming Huang.*  
Contributed to proposal development and writing.

## ■ TEACHING EXPERIENCE

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**Course Instructor MSCI 331: Introduction to Optimization — University of Waterloo**

*May – Aug 2023 Sole instructor, options section; 112 upper-year undergraduates*

- **Student Course Perceptions: 4.29/5 average across the four instructor characteristics** (item range 4.04–4.55; 29 of 112 responding)
- Designed and administered online auto-graded assignments, written quizzes, midterms and final examinations; supervised two teaching assistants
- Held full administrative responsibility for the section, including announcements, grade entry and documentation

**Teaching Assistant Dept. of Management Science & Engineering, University of Waterloo**

*Twelve terms across eight courses, 2017–2023*

- 2019 – 2023 ■ *MSCI 251: Probability & Statistics for Engineers I* — designed and taught the applied R laboratories, and rebuilt them for remote delivery during the pandemic (5 terms; Dr. H. Mahmoudzadeh)
- 2023 ■ *MSCI 734: Network Models* — assignment and project grading (Dr. F. Gzara)
- 2022 ■ *MSCI 435: Advanced Optimization Techniques* — tutorials, grading and student support (Dr. S. Elhedhli)
- 2022 ■ *MSCI 703: Applied Optimization* — grading and solution manuals (Dr. S. Elhedhli)
- 2019 ■ *MSCI 401/402: Design Project* — course logistics and discussion sessions (Dr. A. Hurst)
- 2018 ■ *Work Report Marking* — assessment and written feedback on work-term reports (D. Chan)
- 2017 ■ *MSCI 436: Decision Support Systems* — built and taught weekly Visual Basic tutorials, office hours, grading; 42 students (Dr. M. Karimi)

- 2017 ■ *MSCI 331: Introduction to Optimization* — tutorials, office hours, grading and posted solutions; 41 students (Dr. A. Hiassat)

**Courses Prepared to Teach** Introduction to optimization, and linear and integer programming (taught as sole instructor); probability and statistics for engineers (five terms of applied R laboratories); introductory programming (Python); applied linear algebra; optimization under uncertainty (robust and stochastic); large-scale optimization; simulation; healthcare systems engineering and analytics; decision support systems.

## ■ RESEARCH MENTORSHIP

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**Undergraduate Researchers** **Dept. of Management Science & Engineering, University of Waterloo**  
*Six undergraduate co-op students and research assistants supervised, 2018–2023*

2023 ■ **Gatik Gola** (co-op) — motivated the project, recommended methods and ran weekly meetings for a CT voxel sampling study; *co-author, Medical Physics 2025*

2023 ■ **Karina Wilk** (co-op) — onboarding and weekly planning for the hospital optimization pipeline begun during the Grand River Hospital internship

2021 – 2022 ■ **William Kwong** (research assistant) — proposed and oversaw an interface for running sampling protocols for fluence map optimization; *co-author, Medical Physics 2025*

2021 ■ **Tiantian Li** (research assistant) — exploratory project adapting open-source radiation therapy tools (CERR, matRad) to optimization software

2019 ■ **Gogilan Selvarajah** (research assistant) — pair programming and project direction for a sliding-window IMRT visualization

2018 ■ **Jessie Yeung** (research assistant) — background research and data acquisition for an IMRT sliding-window modelling project

## ■ RESEARCH & CLINICAL EXPERIENCE

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**Graduate Researcher** **Dept. of Management Science & Engineering, University of Waterloo**  
*Sept 2016 – Aug 2024* *Supervised by Dr. Houra Mahmoudzadeh.*

- Built optimization models for radiation therapy treatment planning, tailored to breathing motion uncertainty in breast cancer treatment
- Prototyped and tested models against large clinical datasets using a custom C++/CPLEX code base
- Implemented structured visualizations of plan quality in MATLAB

**Clinical Research Intern** **Medical Physics Department, Grand River Hospital, Kitchener, ON**  
*Oct – Dec 2018* *Supervised by Dr. Ernest Osei.*

- Built the I/O bridge between Varian Eclipse and open-source CERR modules for importing clinical DICOM plans
- Wrote custom collimator rotation code and validated the pipeline on patient plans

**Thesis Student** **Toronto Intelligent Decision Engineering Laboratory, University of Toronto**  
*Sept 2015 – Apr 2016* *Supervised by Dr. J. Christopher Beck.*

- Developed A\*-based batch scheduling algorithms, with C++ implementations of Dijkstra and A\*

**Capstone Student** **Addiction Medicine Services, Centre for Addiction and Mental Health, Toronto, ON**  
*Sept 2015 – Apr 2016* *Supervised by Dr. Michael Carter.*

- Streamlined clinic scheduling and resource allocation, and built a simulation to validate the recommended interventions

- Undergraduate Researcher** **Medical Operations Research Laboratory, University of Toronto**  
*Fall 2012 – May 2014* *Supervised by Dr. Dionne Aleman.*
- Took two interactive radiosurgery planning prototypes from concept to completion in self-taught MATLAB
- Case Study Author** **Industrial Engineering Department, University of Toronto**  
*Fall 2014* *Case adopted as a teaching aid in Cases in Operational Research (MIE367).*
- Wrote a multi-criteria decision-making case around a restaurant venture, with realistic data distributions and a financial planning simulation

## ■ PROFESSIONAL SERVICE & LEADERSHIP

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- Journal Refereeing** Referee for *Computers & Operations Research*, *European Journal of Operational Research*, *International Transactions in Operational Research*, *Medical Physics*, *Operations Research Forum* and *Quantitative Imaging in Medicine and Surgery*
- Conference Service**
- Session Chair, *Data-Driven Optimization in Healthcare*, INFORMS Computing Society (ICS) Conference, Toronto, ON, 2025; also a conference volunteer
  - Session Chair, CORS/INFORMS International Conference, Vancouver, BC, 2022
  - Designed the virtual conference venue (Gather Town) for the CORS Annual Conference, 2021, with Dr. Kiefer Burgess
- Memberships** INFORMS (Health Applications Society); Canadian Operational Research Society (CORS)
- National Student Councillor** **Canadian Operational Research Society (CORS)**  
*2024 – 2025* *Elected to represent the national student body.*
- Led the post-pandemic revival of student chapters across Canada
- President** **The Operations Research Challenge (TORCH), University of Waterloo**  
*Sept 2017 – 2020* *Founded the Waterloo-region edition of the annual high school competition.*
- Led logistics for the 2018, 2019 and 2020\* competitions: student and volunteer recruitment, funding, testing and scheduling
  - Wrote competition problems and validators, and coordinated the automated grading software, with volunteers from Waterloo, Concordia and Toronto
- Student Chapter President** **Canadian Operational Research Society (CORS), Waterloo, ON**  
*Sept 2017 – Sept 2018* *Ran the local student chapter's speaker and social programme.*
- Coordinated speakers and events, onboarding, communications and website oversight
  - Grew active membership from fewer than 10 to nearly 30
- President** **Management Sciences Student Association, University of Waterloo**  
*Sept 2016 – Aug 2017* *Elected departmental representative for graduate students.*
- Budgeted and ran invited speaker and social events; represented students at monthly departmental and cross-departmental meetings
- Executive Member** **The Operations Research Challenge (TORCH), University of Toronto**  
*Fall 2014 – Winter 2016* *Co-Chief Creative Officer (2016); Assistant Editor-in-Chief (2015).*
- Oversaw creation of the 2016 contest, edited the 2015 problem set, and wrote a scheduling problem with an automated solver for grading

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\*Cancelled due to the pandemic

## ■ INDUSTRY EXPERIENCE

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| <b>Rational Software Developer</b><br><i>May 2014 – Aug 2015</i> | <b>IBM Canada Software Lab, Markham, ON</b><br><i>Eclipse-based enterprise software development, testing and debugging (TPF Toolkit, RDi); supervised by Rob Cecco.</i> |
| <b>Engineering Intern</b><br><i>Summer 2012</i>                  | <b>Rotork Controls Canada, Mississauga, ON</b><br><i>Engineering drawings, customer documentation, and a Visual Basic database for intracompany drawing retrieval.</i>  |

## ■ TECHNICAL SKILLS

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|------------------------------|------------------------------------------|
| <b>Languages &amp; Tools</b> | Python, C++, Java, MATLAB, R, $\LaTeX$   |
| <b>Optimization</b>          | CPLEX, Gurobi, HiGHS, AMPL, Excel Solver |
| <b>Clinical / Domain</b>     | Varian Eclipse, CERR, matRad, DICOM-RT   |

*Last updated: October 7, 2026*